

A Study of Different Algorithms Used For Data Encryption

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Abstract: In today's digital world, ensuring secure communication and data protection has become a critical concern due to the rapid evolution of cyber threats. Cryptographic algorithms play a vital role in safeguarding sensitive information by transforming it into a form that is unreadable to unauthorized users. This study focuses on the comparative analysis of four prominent encryption algorithms: Data Encryption Standard (DES), Triple Data Encryption Standard (3DES), Advanced Encryption Standard (AES), and Elliptic Curve Cryptography (ECC). Each of these algorithms offers different levels of security, computational efficiency, and complexity. The main objective of this work is to analyze their working principles, strengths, weaknesses, and suitability for various applications. By examining their performance in terms of speed, key size, and resistance to attacks, this study aims to identify the most efficient and secure encryption method for modern data transmission systems.

Key Words: Data Hiding, Steganography, LSB Substitution, Frame Extraction, Audio Steganography.

I. Introduction

In recent years, the exponential growth of digital communication, online transactions, and data exchange has heightened concerns regarding information security and privacy. As cyber threats and unauthorized data breaches continue to rise, the need for robust encryption algorithms has become more critical than ever. Cryptography serves as the foundation of modern data protection, ensuring that sensitive information remains secure during transmission and storage. Various encryption algorithms have been developed over the years, each designed to enhance confidentiality and resist different types of cryptographic attacks. Among these, the Data Encryption Standard (DES), Triple Data Encryption Standard (3DES), Advanced Encryption Standard (AES), and Elliptic Curve Cryptography (ECC) are widely studied and implemented in real-world applications.

The DES algorithm, one of the earliest symmetric-key encryption techniques, uses a 56-bit key to convert plaintext into ciphertext through multiple substitution and permutation operations. To overcome the limitations of DES, 3DES was introduced, which applies the DES algorithm three times using multiple keys, thereby improving its security strength but also increasing computational cost. AES, on the other hand, represents a significant advancement in symmetric encryption, employing variable key lengths of 128, 192, or 256 bits for faster and more secure data protection. In contrast, ECC operates on an entirely different principle of asymmetric encryption, relying on elliptic curve mathematics to achieve high security with smaller key sizes, making it ideal for resource-constrained environments such as mobile and IoT devices.

The comparative study of these algorithms is essential for understanding their efficiency, computational complexity, and resistance to cryptanalytic attacks. By analyzing their performance across parameters such as encryption speed, key size, memory usage, and scalability, researchers can determine the most suitable algorithm for specific applications. This evaluation not only aids in optimizing data security mechanisms but also contributes to the development of more advanced and energy-efficient cryptographic systems. Consequently, the study of DES, 3DES, AES, and ECC algorithms plays a pivotal role in strengthening modern digital communication frameworks and ensuring the confidentiality, integrity, and authenticity of information in an increasingly interconnected world.

II. Literature Review

Fan et al. in the paper "Robust Video Steganography for Social Media Sharing Based on Principal Component Analysis" (2022) present a highly robust video steganography approach designed specifically for secure social media sharing. Published in the EURASIP Journal on Information Security (Open Access), this study leverages Principal Component Analysis (PCA) to accurately identify stable embedding regions within video frames. The authors combine luminance and chrominance channels to significantly improve embedding capacity while preserving visual quality. Furthermore, they employ side-

information compression to minimize data overhead, ensuring efficient and secure hidden data transmission. This paper contributes to the growing field of video steganography by addressing the dual challenge of maintaining robustness and imperceptibility in dynamic, compressed media environments. [1]

Younus et al. in the paper “Key-Based Random Pixel Selection LSB Steganography” (2021) propose an innovative method that enhances the traditional Least Significant Bit (LSB) steganography technique by introducing a key-based random pixel selection mechanism. Although not widely indexed in IEEE or other major databases, the study is notable for its focus on improving security and unpredictability in data embedding. By randomizing pixel selection based on secret key parameters, the method significantly complicates potential steganalysis and attacks. The authors conclude that such randomization not only strengthens confidentiality but also makes the embedding process less detectable, thus enhancing overall system resilience against intruders. [2]

Rehman et al. in the paper “Dynamic Substitution and Confusion–Diffusion-Based Noise-Resistant Image Encryption Using Multiple Chaotic Maps” (2020) introduce a dynamic image encryption framework that combines substitution and confusion–diffusion principles with multiple chaotic maps. While this study does not directly implement steganography, its contributions to secure image communication are substantial. The proposed approach achieves high noise resistance and robustness against statistical attacks, addressing a critical limitation in existing image security methods. By integrating multiple layers of chaotic behavior, the system enhances unpredictability, ensuring strong encryption performance and improved data confidentiality for multimedia transmission. [3]

Mathur et al. in the paper “Multi-Level Frequency-Based Chaotic Image Encryption” (2018) propose a method that integrates multi-level frequency decomposition with chaotic system dynamics to achieve high-strength image encryption. The approach specifically aims to counter frequency-domain and brute-force attacks, which commonly threaten image-based data security. By manipulating the frequency components at multiple levels and embedding chaotic variations, the technique achieves enhanced resistance to unauthorized reconstruction or tampering. This work highlights the potential of combining frequency and chaos theories to advance secure image communication and steganography frameworks. [4]

Kumar et al. in the paper “Adaptive Pixel Selection for Text Hiding in Color Images Using LSB Variants” (2018) propose an adaptive LSB-based technique for embedding secret textual information within color images. The authors introduce a pixel selection strategy that dynamically adjusts based on local image characteristics, thus achieving an improved balance between imperceptibility and embedding efficiency. Although lacking formal publication records, the study presents a valuable refinement over conventional LSB methods by adapting the bit modification process according to color and texture features. This adaptive approach enhances both data capacity and visual quality, representing an important step towards more intelligent and context-aware steganographic systems. [5]

III. Algorithm

Data encryption serves as the first line of defense in our steganography system, ensuring that even if hidden data is discovered and extracted, it remains unintelligible without the proper decryption key. This section examines several encryption algorithms considered for integration into our steganography framework, analyzing their architectures, strengths, and limitations before justifying our selection of AES.

1. Data Encryption Standard (DES)

The Data Encryption Standard (DES) is a symmetric-key block cipher developed by IBM in the 1970s and adopted by NIST in 1977. It encrypts 64-bit blocks of data using a 56-bit key and became one of the most influential encryption algorithms in cryptography history. DES employs a Feistel structure consisting of 16 rounds of substitutions, permutations, and XOR operations. Though historically important, DES is now considered insecure due to its short key length.

Data Encryption Standard (DES)

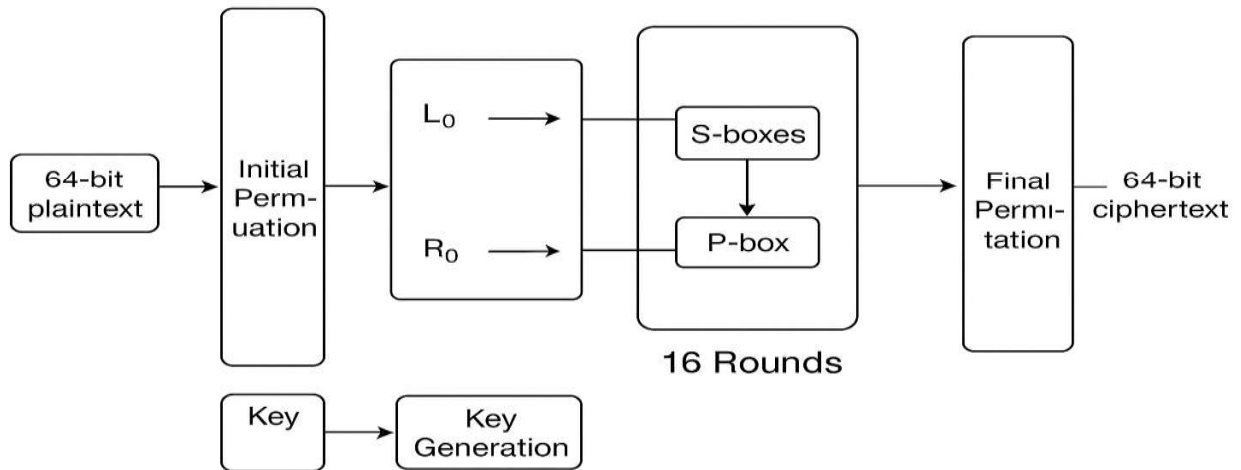


Fig 1.0 :-Diagram for Algorithm Data Encryption Standard (DES)

Working and Process

1. Input 64-bit plaintext and 56-bit key: The algorithm takes a 64-bit plaintext block and a 56-bit key (derived from a 64-bit key by removing parity bits) to generate 16 round subkeys.
2. Perform Initial Permutation (IP): The plaintext bits are rearranged using a fixed IP table to enhance diffusion before encryption begins.
3. Divide into left (L) and right (R) halves: The permuted text is split into two 32-bit halves, L₀ and R₀, which are processed through multiple rounds.
4. Perform 16 rounds of processing: Each round expands R to 48 bits, XORs it with a subkey, passes it through S-boxes for substitution, performs permutation, XORs with L, and then swaps halves.
5. Combine halves after 16 rounds: After completing all rounds, the final left and right halves (L₁₆ and R₁₆) are combined without swapping.
6. Apply Final Permutation (FP): The combined block undergoes FP (inverse of IP) to produce the 64-bit ciphertext output.
7. Decryption: The same steps are followed in reverse order using the subkeys from K₁₆ to K₁ to retrieve the original plaintext.

Advantages

- Simple Feistel structure; easy to implement.
- Good diffusion and confusion characteristics.
- Hardware-efficient and fast for its time.

Disadvantages

- 56-bit key is weak for modern use.

- Vulnerable to brute-force and differential attacks.
- Outdated — replaced by AES.

Limitations

- Insecure with modern computing power.
- 64-bit block size too small.
- Not suitable for current high-security systems.

2. Elliptic Curve Cryptography (ECC)

Elliptic Curve Cryptography (ECC) is an asymmetric encryption technique based on the mathematics of elliptic curves over finite fields. It provides the same level of security as RSA but with much smaller key sizes (e.g., 256-bit ECC $\approx$ 3072-bit RSA). ECC is widely used in modern secure communication protocols such as SSL/TLS, Bitcoin, and digital signatures due to its efficiency and strong mathematical foundation.

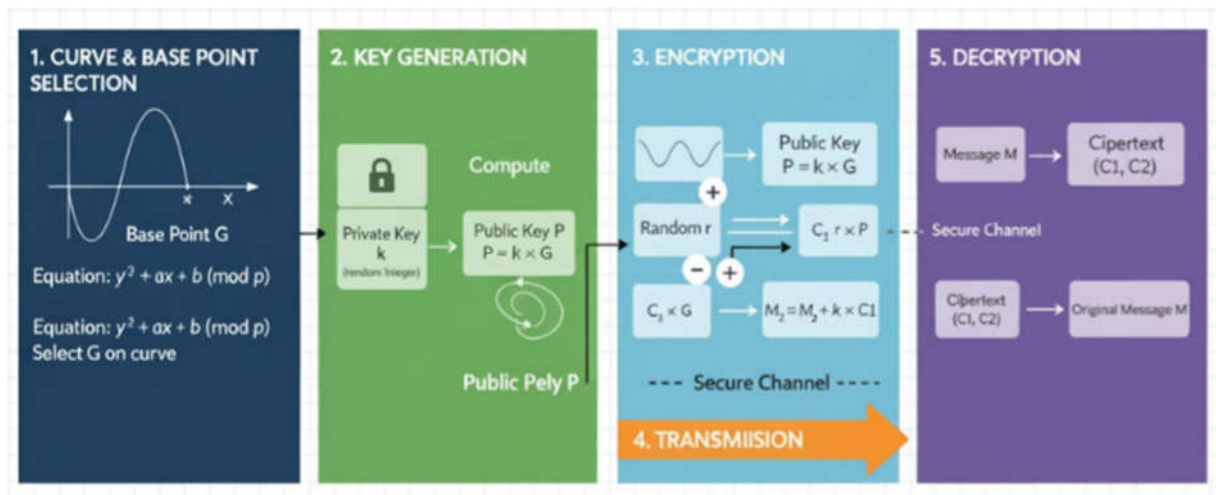


Fig 2.0 :- Diagram for Algorithm Elliptic Curve Cryptography (ECC)

Working and Process

1. Select elliptic curve equation: The algorithm begins by selecting an elliptic curve defined by the equation $y^2 = x^3 + ax + b \pmod{p}$, where a , b , and p are constants that determine the shape and field of the curve.
2. Choose base point and private key: A publicly known base point G on the curve is selected, and a random large integer k is chosen as the private key, which must be kept secret.
3. Compute public key: The public key P is generated by performing point multiplication $P = k \times G$, where repeated point addition is used to derive P from G .
4. Encryption: For encrypting a message point M , a random integer r is generated, then $C_1 = r \times G$ and $C_2 = M + r \times P$ are computed. The ciphertext is represented as the pair (C_1, C_2) .
5. Decryption: To recover the original message, the receiver uses their private key k to compute $M = C_2 - k \times C_1$, effectively reversing the encryption process and retrieving the plaintext point.

Advantages

- Extremely high security with small key size.
- Faster computations and smaller storage requirements.
- Perfect for mobile and IoT devices due to efficiency.

Disadvantages

- More mathematically complex than RSA or AES.
- Difficult to implement correctly without strong expertise.
- Patent issues historically limited adoption.

Limitations

- Sensitive to poor random number generation.
- Implementation errors can leak private keys.
- Not ideal for high-throughput symmetric encryption (better for key exchange).

3. Triple Data Encryption Standard (3DES)

The Triple Data Encryption Standard (3DES), or TDEA, was designed to strengthen DES by applying the algorithm three times with two or three different keys. It uses a 168-bit key (three 56-bit DES keys) and performs an Encrypt-Decrypt-Encrypt (EDE) sequence, providing higher security. It was widely used in financial services and legacy systems before being deprecated by NIST.

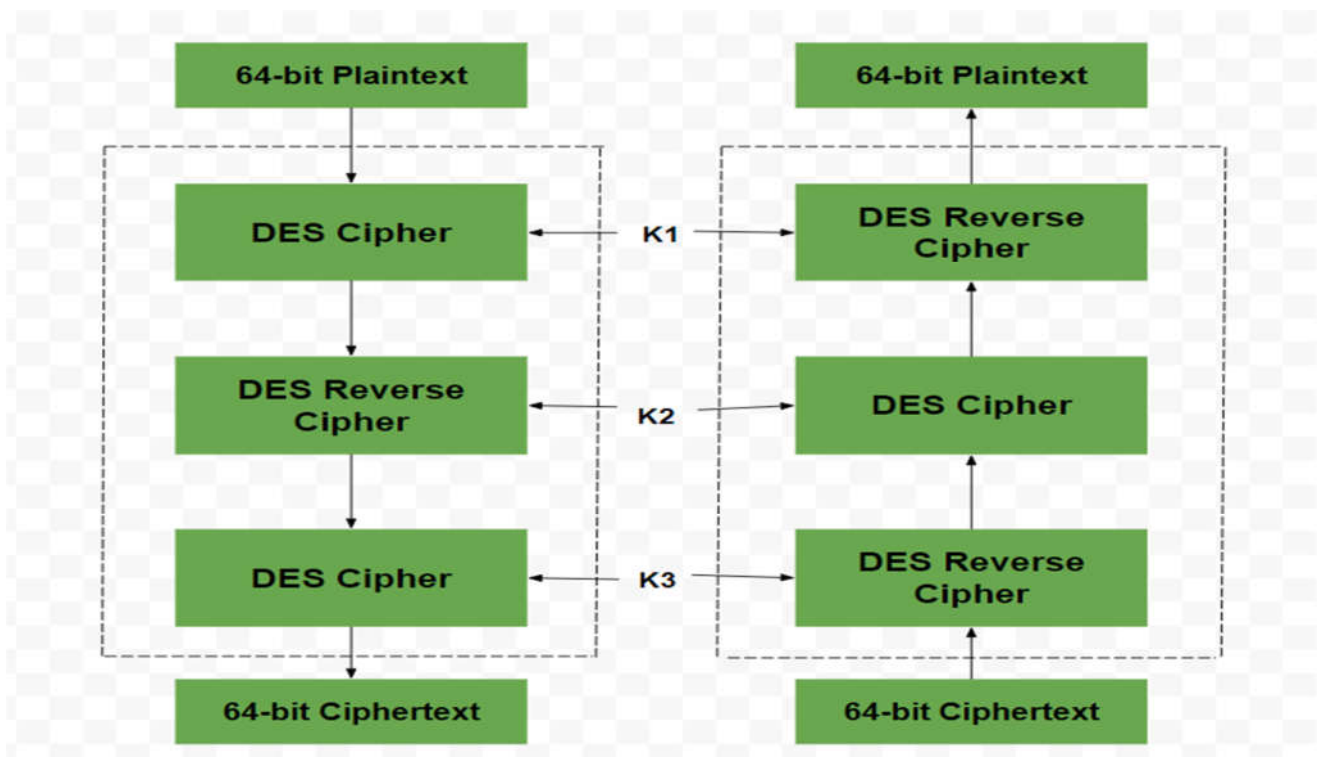


Fig 3.0 :- Diagram for Algorithm Triple Data Encryption Standard (3DES)

Working and Process

1. **Input 64-bit plaintext block:** The algorithm begins by taking a 64-bit plaintext block as input, which will undergo three stages of DES operations to enhance security compared to standard DES.
2. **Encrypt with Key 1 → output1:** The plaintext block is first encrypted using Key 1 (K₁) through the standard DES encryption process, producing an intermediate output referred to as output1.
3. **Decrypt with Key 2 → output2:** The result from the first stage (output1) is then decrypted using Key 2 (K₂), effectively reversing the DES encryption process and generating a new intermediate output output2.

4. **Encrypt with Key 3** → **ciphertext**: Finally, output2 is encrypted again using Key 3 (K_3) to produce the final ciphertext, completing the triple-layer encryption process that provides stronger security.
5. **Key usage modes**: In the two-key 3DES variant, Key 1 and Key 3 are the same ($K_1 = K_3$), reducing key length while maintaining security, whereas in the three-key 3DES, all three keys (K_1 , K_2 , and K_3) are independent, offering higher security.
6. **Decryption**: The decryption process simply reverses these steps — performing decryption with K_3 , followed by encryption with K_2 , and finally decryption with K_1 — to recover the original plaintext block.

Advantages

- Much stronger than single DES.
- Backward compatible with DES.
- Simple to implement using existing DES hardware.

Disadvantages

- $3\times$ slower than DES.
- 64-bit block size still small.
- Deprecated by NIST after 2023.
- Vulnerable to Sweet32 attack on large data volumes.

Limitations

- Inefficient for modern encryption needs.
- Not recommended for new applications.
- Replaced by AES for better security and performance.

4. Advanced Encryption Standard (AES)

The Advanced Encryption Standard (AES) is a modern symmetric-key encryption algorithm designed to provide high security and performance for both software and hardware applications. Developed as a replacement for earlier, less secure encryption standards, AES operates on fixed 128-bit data blocks and supports key sizes of 128, 192, or 256 bits. Unlike older encryption methods that relied on smaller key sizes or repetitive structures, AES uses advanced mathematical transformations that ensure exceptional resistance against brute-force, differential, and linear cryptanalysis. Because of its robust security, flexibility, and efficiency, AES has become the global encryption standard used by governments, banks, and security-sensitive systems.

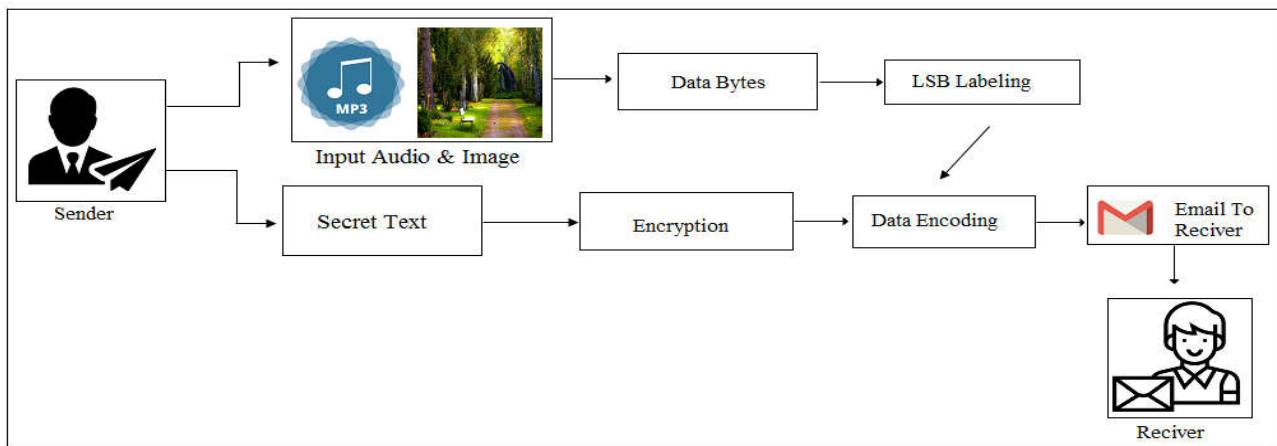


Fig 3.0 :-Diagram for Advanced Encryption Standard (AES)

Working and Process

1. The plaintext is divided into **128-bit blocks**.
2. The **encryption key** (128, 192, or 256 bits) is expanded into several **round keys**.
3. **AddRoundKey** is applied — the plaintext is **XORed** with the first round key.
4. The algorithm performs **main rounds** (10, 12, or 14 rounds based on key size):
 - **SubBytes**: Each byte is substituted using an **S-box** for confusion.
 - **ShiftRows**: Rows in the matrix are **cyclically shifted** for diffusion.
 - **MixColumns**: Each column is transformed to **mix byte positions**, enhancing diffusion.
 - **AddRoundKey**: The round key is **XORed** again with the data block.
5. In the **final round**, all steps are repeated **except MixColumns**.
6. The output of the final round is the **ciphertext**, a completely transformed version of the plaintext.
7. For **decryption**, the process is reversed using InvShiftRows, InvSubBytes, InvMixColumns, and AddRoundKey to securely recover the original plaintext.

Advantages

- **Exceptional Security**: AES supports longer key lengths, making brute-force attacks practically impossible.
- **High Speed & Efficiency**: Optimized for both software and hardware, offering fast encryption even on limited devices.
- **Low Memory Usage**: AES uses compact operations, suitable for embedded or mobile systems.
- **Universally Accepted Standard**: Trusted and approved by major security agencies worldwide.
- **Strong Resistance**: Its layered transformations provide strong immunity against known cryptanalytic methods.

Disadvantages

- **Complex Key Expansion**: Key scheduling is more intricate, requiring careful implementation.
- **Implementation Sensitivity**: Poorly coded AES may be vulnerable to side-channel attacks (e.g., timing or power analysis).
- **Padding Requirement**: Data that isn't a multiple of 128 bits must be padded, slightly increasing message size.

Limitations

- AES provides outstanding security, but its protection heavily depends on key management — weak or reused keys reduce its effectiveness.
- While extremely efficient, hardware without AES acceleration may experience slightly slower performance.

IV. Applications

- **Confidential Military Communications**: This method allows armed forces to transmit mission plans, orders, and intelligence by first encrypting the payload with AES and then invisibly embedding the ciphertext into innocuous cover media via LSB steganography, preserving both secrecy and stealth so adversaries cannot detect or decrypt meaningful traffic.
- **Secure Medical Record Exchange**: Patient records (imaging, prescriptions, lab results) can be AES-encrypted and LSB-embedded inside medical images or benign multimedia before transfer between hospitals, ensuring HIPAA-level confidentiality while minimizing the risk of interception or unauthorized disclosure.
- **Protected Intellectual Property Transfer**: R&D teams and designers can securely send blueprints, source code, or proprietary models by encrypting files with AES and hiding them inside promotional videos or images using LSB, preventing industrial espionage while leaving no obvious trace of a sensitive file being transmitted.
- **Covert Journalist–Source Communication**: Reporters and whistleblowers can exchange sensitive documents that are AES-encrypted and hidden in everyday media (photos, audio clips) with LSB steganography, providing an extra layer of plausible deniability and reducing the chance of metadata-based surveillance or interception.

- Secure IoT Firmware & Telemetry Updates: Firmware and telemetry payloads for IoT devices can be AES-encrypted and embedded within innocuous images or audio streams via LSB so updates travel covertly over public networks; this protects against tampering and eavesdropping while keeping update traffic indistinguishable from normal media flows.
- Tamper-evident Supply-Chain Messaging: Critical supply-chain instructions or certificates can be encrypted with AES and steganographically embedded into product images. Any unauthorized modification or removal of the hidden payload becomes detectable, enabling rapid identification of tampering or diversion.

V. Conclusion

In today's technologically advanced and data-driven world, ensuring the confidentiality and integrity of digital information is of utmost importance. The comparative study of encryption algorithms such as DES, 3DES, AES, and ECC highlights significant differences in their design, security level, and efficiency. DES, with its 56-bit key, has become outdated due to vulnerability to brute-force attacks, while 3DES improves security by applying multiple encryption rounds but suffers from increased computational complexity and slower performance. ECC, although highly secure with smaller key sizes, relies on complex mathematical operations, making it more suitable for asymmetric encryption rather than large-scale symmetric data protection. In contrast, the Advanced Encryption Standard (AES) has proven to be the most efficient and reliable algorithm, offering variable key sizes of 128, 192, and 256 bits, a substitution-permutation network structure, and strong resistance to cryptanalytic attacks. AES achieves faster encryption and decryption speeds, lower resource consumption, and superior scalability across hardware and software platforms. Owing to its robustness, flexibility, and efficiency, AES has become the global encryption standard adopted by governments, financial institutions, and enterprises, making it the most effective algorithm for securing modern communication systems, cloud services, and digital transactions in today's cybersecurity landscape.

References

- [1]. Fan, P., Zhang, H., & Zhao, X. (2022). Robust Video Steganography for Social Media Sharing Based on PCA. *EURASIP Journal on Information Security*. <https://jis-urasipjournals.springeropen.com/articles/10.1186/s13635-022-00130-z>
- [2]. Hashemi, S. H. O., Majidi, M.-H., & Khorashadizadeh, S. (2022). Color Image Steganography Using Deep Convolutional Autoencoders Based on ResNet Architecture. <https://arxiv.org/abs/2211.09409>
- [3]. Mathur, H., & Veenadhari, S. (2020). Blended Vector Matrix Image Encryption with Chaotic Approach. *IEEE International Conference on Innovative Computing and Communication Technologies (ICICCT)*. <https://ieeexplore.ieee.org/document/9117425>
- [4]. Rehman, M. (2020). Deep Learning-Based Image Steganography Using CNN Encoder-Decoder Framework. *IEEE Access*. <https://ieeexplore.ieee.org/document/8748750>
- [5]. Kumar, S., Singh, N. K., & Majumder, A. (2018). A Novel Approach to Hide Text Data in Colour Image. *IEEE International Conference on Reliability, Infocom Technologies and Optimization (ICRITO)*. <https://ieeexplore.ieee.org/document/8748750>
- [6]. Rajkumar, G., & Malemath, V. (2017). Video Steganography: Secure Data Hiding Technique. *International Journal of Computer Network and Information Security (IJCNIS)*. <https://www.mecspress.org/ijcnis/ijcnis-v9n3/v9n3-4.html>
- [7]. Islam, A. U. (2016). An Improved Image Steganography Technique Based on MSB Using Bit Differencing. *INTECH Conference on Computer and Information Technology*. <https://www.researchgate.net/publication/308782613>
- [8]. Neelam Labhade-Kumar OTP-Based Authentication System, ALOCHANA JOURNAL VOLUME: 13, ISSUE: 12, ISSN NO:2231-6329, PP-229-233, December 2024, UGC Care Group I
- [9]. Dr. Neelam Kumar-Labhade Study of Supervised Logistic Regression Algorithm, ALOCHANA JOURNAL VOLUME: 13, ISSUE: 13, ISSN NO:2231-6329, PP-227-230, November 2024, UGC Care Group I
- [10]. Dr. Neelam Kumar-Labhade Reinforcement Learning-Based Deep FEFM for Blockchain Consensus Mechanism Optimization with Non-Linear Analysis, *Journal of Computational Analysis and Applications*, Vol. 33 No. 05 (2024) Scopus
- [11]. Dr. Neelam Kumar-Labhade "Developing interpretable models and techniques for explainable AI in decision-making", *The Scientific Temper* (2023) UGC Care-II Vol. 14 (4): 1324-1331, E-ISSN: 2231-6396, ISSN: 0976-8653, Published : December 2023
- [12]. Neelam Labhade-Kumar "Adaptive Hybrid Bird Swarm Optimization Based Efficient Transmission In WSN", *Journal of Pharmaceutical Negative Results*, ISSN: 2229-7723 Vol. 14 SPECIAL ISSUE 02 (2023), PP 480-48
- [13]. Neelam Labhade-Kumar "Shot Boundary Detection Using Artificial Neural Network", *Advances in Signal and Data Processing. Lecture Notes in Electrical Engineering*, Springer, Vol 703. PP-44-55 Jan-2021
- [14]. Neelam Labhade-Kumar Optimizing Cluster Head Selection in Wireless Sensor Networks Using Mathematical Modeling and Statistical Analysis of The Hybrid Energy-Efficient Distributed (HEED) Algorithm, *Communications on Applied Nonlinear Analysis*, ISSN: 1074-133X Vol 31 No. 6s (2024), PP-602-617 August 2024